

Conversion Referencer

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Project Description:

My project website for this course is called Conversion Referencer. The goal of the website is to allow people/user to get a quick conversion of a particular measurement they are looking for. For example, oftentimes, you may find yourself making a recipe from online, and it calls for 68 ounces of flour, but you might not know how many cups that is, so this is where the site comes in handy. I love cooking and hate having to look up measurement conversions every time I find a new recipe; that is how I got the idea. The website includes a dark and light mode, 6 categories of measurements (cooking, construction, health, tech, time, and travel), a form to suggest new measurements, and lastly a custom converter for more specific measurements in mind.

Reflection:

Difficulties encountered:

1. My first idea and hurdle was adding JavaScript to make light/dark modes.
2. Setting up media queries for smaller-width screens (mobile devices).
3. Getting my suggestion form to be correctly styled the way I wanted, and to then getting it functioning properly.
4. The first custom conversion I had to do, which was the construction category. I had a hard time designing the scope and getting the math correct.

How difficulties were solved:

1. After having the idea of working with light and dark modes, luckily, my CS course (343) started to work with JS and DOM manipulation, so later in the week, I figured it out. Also, I have been working on my own personal website on the side, which was also starting to use the same concept. The hard part was mostly changing and adding all the new styles since my original static website was already in "dark mode."
2. Setting up media queries was more just trial and error until I got it to work, lots of incremental changes to test what works and looks best.
3. I looked up what good forms looked like on Google images and then looked up later how to make use of the user inputs and use mailTo, but my implementation is still fiddly.
4. I used my free time during Thanksgiving Break to work on this. It took much longer than I thought. I pulled up Google's converter and made my look similar to it (used as

inspiration). I looked up online it was common to use units instead of making individual calculations with a bunch of if statements. This breakthrough really help get me rolling.

What could be improved:

- Later on, I want to add more categories and measurements; some of the categories seem to have received less attention from me.
- I want to reduce my number of custom conversion JavaScript files since most of them are identical; the only thing that changes is the specified units. I started off not knowing how the math would be implemented so I originally had the code all separated.
- The suggestion form could use more polish. I might need to try something other than mailTo; maybe Google Forms.
- Later on, I could add multiple languages, as I just worked with that implementation for my year-long project in my CS 343 course.
- Use math.units for custom conversions instead of the method I used. I needed to do more research before working on the version I made.

Resources used:

- Inspiration from Google Images for the suggestion form and Google's conversion table look for my own.
- For custom scrollbar in header:
https://www.w3schools.com/howto/howto_css_custom_scrollbar.asp
- Idea for easier custom conversions:
<https://stackoverflow.com/questions/58187907/how-to-convert-imperial-to-metric-units>
- mailTo form idea, since I am limited without using a web server:
 - https://www.reddit.com/r/webdev/comments/14gcq8l/are_there_any_free_and_safe_form_handlers_i_can/
 - <https://forum.freecodecamp.org/t/a-form-that-can-send-emails-without-a-backend/387841>